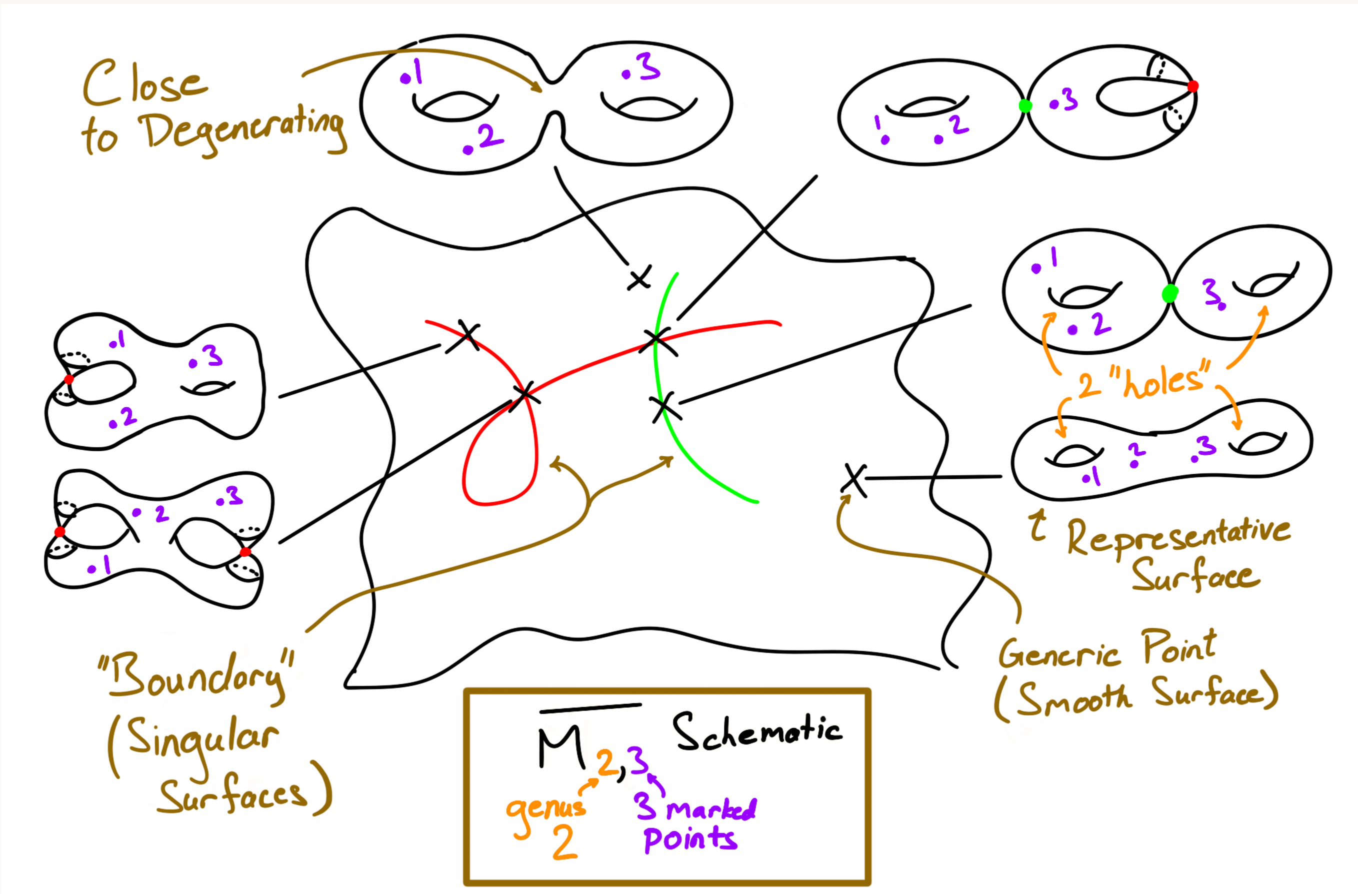
On a Conjecture on the Values of ψ-Correlators

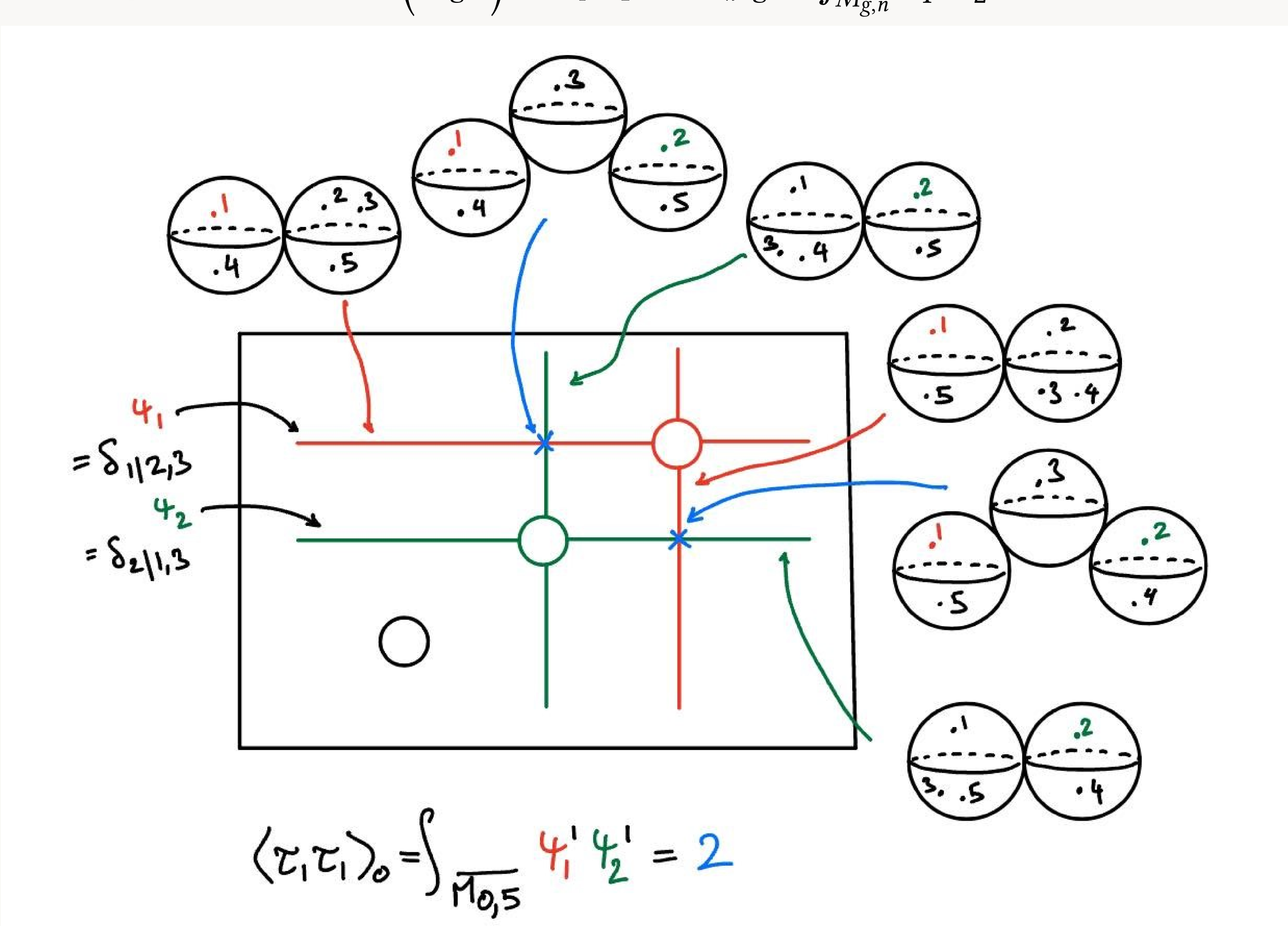
Moduli Spaces

- $\overline{M_{g,n}}$ is a moduli space (parameter space) where each point represents a curve of genus g with n marked points
- The space is compactified with points corresponding to (stable) curves with (nodal) singularities
- The dimension of $\overline{M_{g,n}}$ is $3g-3+n$ and nearby points represent "similar looking curves"



Cohomology, ψ-Classes, and Enumerative Geometry

- Cohomology describes global aspects of geometry using algebra/calculus - The elements of cohomology are subvarieties (embedded geometric objects) and their product corresponds to intersections
- ψ-Classes are natural codimension 1 classes that arise in the cohomology of $\overline{M_{g,n}}$ - For each marked point p_i, there is a corresponding ψ_i
- *ψ-Correlators* are intersection products of ψ-classes: if $\sum d_i = 3g-3+n = \dim\left(\overline{M_{g,n}}\right)$, $\langle \tau_{d_1}\tau_{d_2}\dots\tau_{d_n}\rangle_g = \int_{\overline{M_{g,n}}} \psi_1^{d_1}\psi_2^{d_2}\dots\psi_n^{d_n}$



- Intersection numbers prove to be useful in answering enumerative geometry problems and elucidating new structure, for example:
  - The ELSV formula counts Hurwitz numbers i.e. branched coverings with ramification $\mu = (\mu_1, \dots, \mu_n)$ over ∞ and simple ramification elsewhere - ELSV proves polynomiality of Hurwitz numbers in μ_i, which is hard to do with other methods
  - Kontsevich's formula counts the number of degree d curves passing through $3d-1$ points in the plane - Up to the 19th century, it was only known for $d \le 4$, but Kontsevich's formula remarkably answers it for all d by a recursive relation only needing the simple fact "two points uniquely determine a line"

Overall Goal

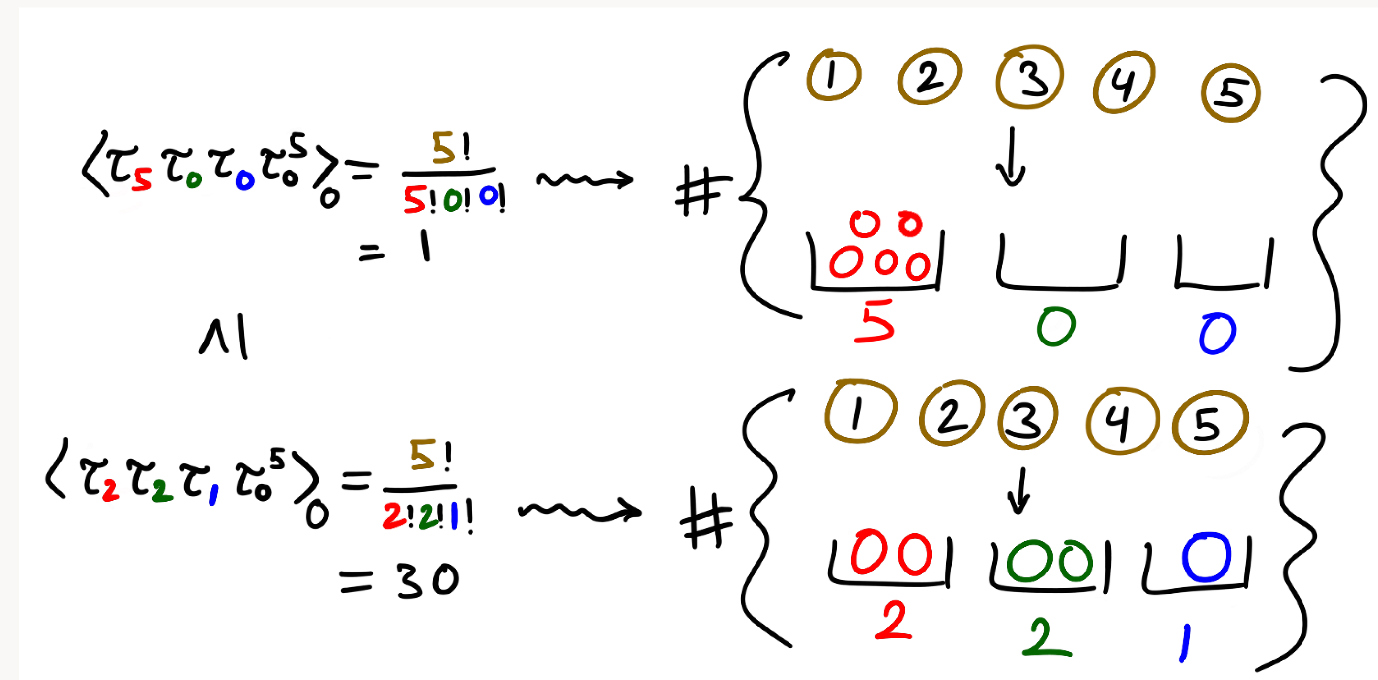
- Understanding the arithmetic of *ψ-Correlators* can give deep insights into the geometry of $\overline{M_{g,n}}$, and hence certain enumerative problems
- While there are recursive formulas, an explicit formula is deemed intractible, so qualitative statements are highly sought after

The goal of this project is to prove the following

Conjecture (Liu, Xu [1])

If $\sum d_i = 3g-3+n$ and $d_1 < d_2$, $\langle\tau_{d_1}\tau_{d_2}\cdots\tau_{d_n}\rangle_g \le \langle\tau_{d_1+1}\tau_{d_2-1}\cdots\tau_{d_n}\rangle_g$ i.e. the value of the ψ-correlators grow as the indices become equidistributed

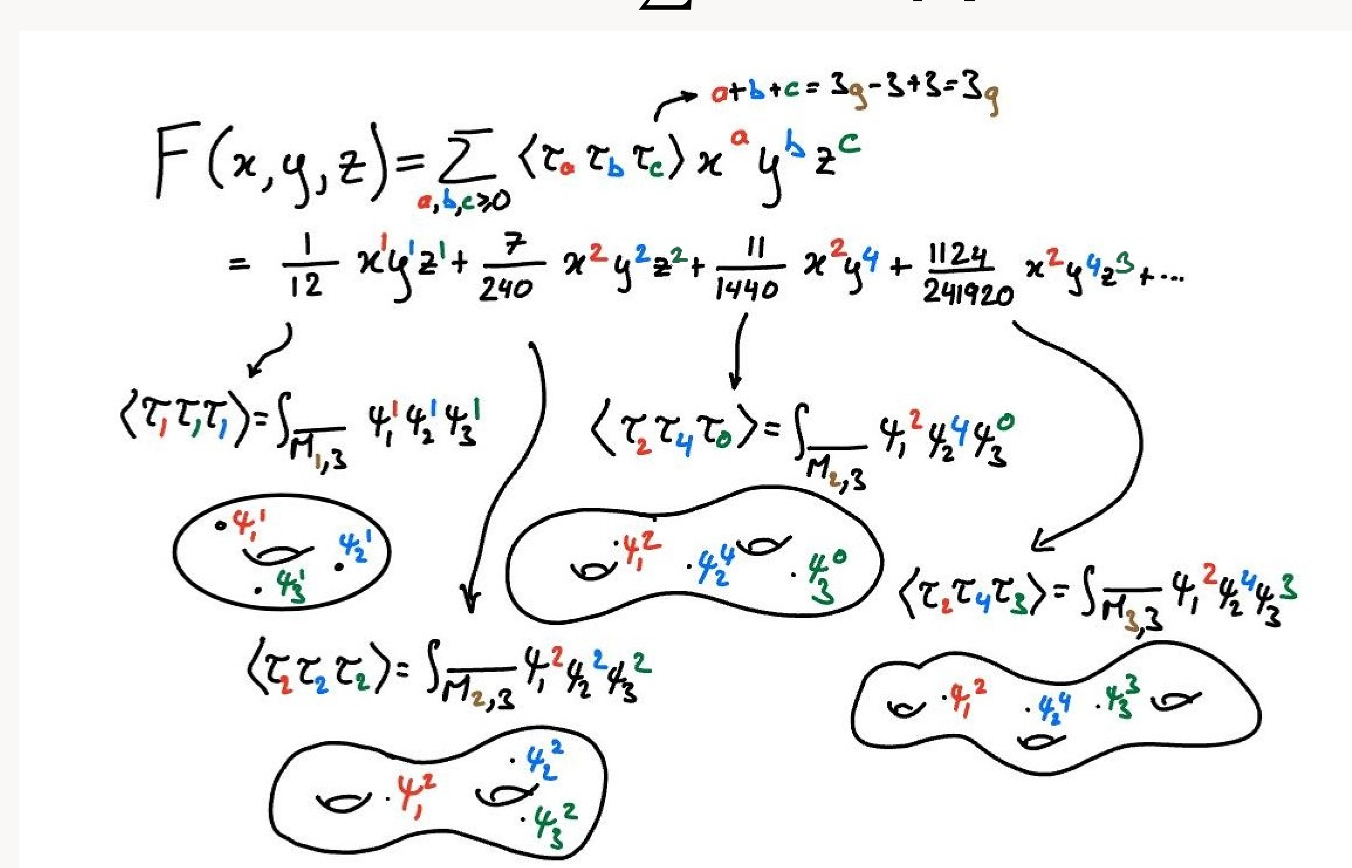
Some reason to believe in this conjecture is that the genus 0 case corresponds to multinomial coefficients which exhibit this behavior



Generating Functions

- An essential component in the computation of ψ-classes is the generating function
- Generating functions package numerical data into functions which allow for algebraic and analytic methods of analyzing them
- The main tool is the n-point function classifying n-point correlators:

$$F(x_1,\dots,x_n) = \sum \langle\tau_{d_1}\dots\tau_{d_n}\rangle x_1^{d_1}x_2^{d_2}\dots x_n^{d_n}$$



- Liu and Xu [2] have recursive formulas for n-point functions with explicit equations for 2- and 3-point functions allowing for computation

Methods and Results

- The theorem was proved for all 2-correlators - Using numerical estimates on an explicit formula of Zograf [3], it was shown that $\langle\tau_{a+1}\tau_{b-1}\rangle - \langle\tau_a\tau_b\rangle > 0$ when $a < b$
- For three correlators, the related generating function of Zagier [2], $G(x,y,z) = \exp\left(-\frac{x^3+y^3+z^3}{24}\right)F(x,y,z)$, was analyzed and the following alternative formula was found:

$$G(x,y,z) = \sum_{\substack{g,n\in\mathbb{Z}^{\geq 0}\\ s\in\{0,\dots,g\}}} \frac{z^n}{n!}C_{g,s}(xy(x+y))^{g-n}\left(\sum_{k=0}^{\lfloor\frac{n-1}{2}\rfloor} D_{n,k}s^{\underline{n-k}}(x+y)^{2(n-k)}(xy)^k + n!(-xy)^n\right)$$

$$C_{g,s} = \left(2^{g+2s-1}\binom{2(g-s)}{g-s}(2(g-s)+1)\right)^{-1},\ D_{n,k} = \frac{n!}{k!(n-2k-1)!(n-k)},$$
$$\text{and } s^{\underline{r}} = s(s-1)\dots(s-(r-1))$$

- By comparing binomial coefficients, a weaker analog of the conjecture was shown for $G(x,y,z)$ - In most cases, the coefficient of $x^{a+1}y^{b-1}z^n$ in G is at least that of $x^a y^b z^n$ exactly when $a < b$
- It is desirable to understand the geometric relationship between G and F to deduce the main conjecture for 3 correlators using the coefficients of G - The exponential factor suggests some connection with genus 1 components
- This method seems generalizable to higher correlators through the recursive formulas for $G(x_1,\dots,x_n) = \exp\left(-\frac{1}{24}\sum x_i^3\right)F(x_1,\dots,x_n)$ of Liu and Xu [2]

Use of Code

- admcycles [4] is a Sage package which recursively computes ψ-correlators - It was very useful in checking maipulations of expressions like Zograf's formula, and in formulating and checking numerical conjectures
- Sage's symbolic differentiation was very useful in manipulating the 3-point generating function $G(x,y,z)$ to find patterns and conjecture a general form

```
In [1]:  #admcycles to test conjectures
         from admcycles import *
         for i in range(2,5):
             print((8-i,i))
             print(psi_correlator(9-i,i,0,0,0,0)/(4*(2*(9-i)+3))
                   -3*psi_correlator(9-i,i-1)+psi_correlator(9-i+1,i-2))

         (6, 2)
         7/51840
         (5, 3)
         5/20736
         (4, 4)
         14393/37739520

In [2]:  #symbolic differentiation to find patterns in generating functions
         S(x,y,z,r) = ((x*y)^r*(x+y)^(r+1)+(y*z)^r*(y+z)^(r+1)+(x*z)^r*(x+z)^(r+1))/(x+y+z)
         D(x,y,z) = (x+y+z)^3/3-(x^3+y^3+z^3)/3
         T(x,y,z,r,s) = S*D^s
         Tzz=T.diff(z).diff(z)
         latex(factor(Tzz(x,y,0)))

Out[2]:  \frac{{\left(s^{2} x^{4} + 4 \, s^{2} x^{3} y + 6 \, s^{2} x^{2} y^{2} + 4 \, s^{2} x y^{3} + s^{2} y^{4} - s x^{4} - 4 \, s
         x^{3} y - 6 \, s x^{2} y^{2} - 4 \, s x y^{3} - s y^{4} + 2 \, x^{2} y^{2}\right)} \left({\left(x + y\right)} x y\right)^{s}
         \left(x y\right)^{r} {\left(x + y\right)}^{r - 2}}{x^{2} y^{2}}
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Citations

- Liu, K. & Xu, H. (2009) *Intersection Numbers and Automorphisms of Stable Curves.* Michigan Mathematical Journal
- Liu, K. & Xu, H. (2009) *The N-Point Functions for Intersection Numbers on Moduli Spaces of Curves.* Adv. Theor. Math. Phys.
- Zograf, P. (2020) *An Explicit Formula For Witten's 2-Correlators.* Journal of Mathematical Sciences
- Delecroix, V., Schmitt, J., & van Zelm, J. (2021) *admcycles - a Sage package for calculations in the tautological ring of the moduli space of stable curves.* Journal of Software for Algebra and Geometry

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